Physics 215C Homework #1 Solutions

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Shankar 20.1.1

Derive the continuity equation

$$\frac{\partial P}{\partial t} + \nabla \cdot j = 0$$

where $P = \psi^\dagger \psi$ and $j = c\psi^\dagger \alpha \psi$

The Dirac equation takes the equivalent forms

$$i\hbar \frac{\partial |\psi\rangle}{\partial t} = \left(c\alpha \cdot P + \beta mc^2\right)|\psi\rangle$$

$$\frac{\partial \psi}{\partial t} = \left(-c\alpha \cdot \nabla - \frac{i}{\hbar}\beta mc^2\right)\psi$$

The conjugate equation is

$$\frac{\partial \psi^\dagger}{\partial t} = \left(-c\alpha \cdot \nabla + \frac{i}{\hbar}\beta mc^2\right)\psi^\dagger$$

$$\begin{aligned}\frac{\partial P}{\partial t} &= \psi^\dagger \frac{\partial \psi}{\partial t} + \frac{\partial \psi^\dagger}{\partial t}\psi \\ &= -c\left(\psi^\dagger \alpha \cdot \nabla \psi + (\alpha \cdot \nabla \psi^\dagger)\psi\right)\end{aligned}$$

$$\nabla \cdot j = c\left(\nabla \psi^\dagger \alpha \psi + \psi^\dagger \alpha \cdot \nabla \psi\right)$$

Adding both equations together yields the desired result

$$\frac{\partial P}{\partial t} + \nabla \cdot j = 0$$

Both terms in the Hamiltonian, $c\alpha \cdot P$ and βmc^2 are Hermitian, so where did the relative minus sign come from? To show that P is a Hermitian operator you need to integrate by parts. If you are working with the L^2 inner product you can drop boundary terms, but when working locally to derive the continuity equation we don't integrate by parts and get a relative minus sign.

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Show that the probability current j of the previous exercise reduces in the non-relativistic limit to Eq.(5.3.8) [which is the same as Sakurai Eq.(2.4.16)].

The probability current $j = c\psi^\dagger \alpha \psi$ and $\alpha = \begin{pmatrix} 0 & \sigma \\ \sigma & 0 \end{pmatrix}$ We can write the wave function $\psi = \begin{pmatrix} \chi \\ \Phi \end{pmatrix}$ in terms of its relativistic and non-relativistic components, Φ and χ respectively.

$$j = c \begin{pmatrix} \chi^\dagger & \Phi^\dagger \end{pmatrix} \begin{pmatrix} 0 & \sigma \\ \sigma & 0 \end{pmatrix} \begin{pmatrix} \chi \\ \Phi \end{pmatrix} \tag{0.1}$$

$$= c \left(\chi^\dagger \sigma \Phi + \Phi^\dagger \sigma \chi \right) \tag{0.2}$$

In the non-relativistic limit (20.2.13)

$$\Phi \approx \frac{\sigma \cdot \pi}{2mc}$$

$$j = \chi^\dagger \frac{\hat{p}}{2m} \chi + (\frac{\hat{p}}{2m} \chi^\dagger) \chi$$

which is the non-relativistic current (5.3.8)

$$j = \frac{\hbar}{2mi} \left(\psi^* \nabla \psi - \psi \nabla \psi^* \right)$$

from the identification $\hat{p} = -i\hbar\nabla$.

Shankar 20.2.1

Show that

$$\pi \times \pi = \frac{iq\hbar}{c} B$$

where $\pi = P - \frac{qA}{c}$.

$$\begin{aligned} \pi \times \pi &= P \times P - \frac{qA}{c} \times P - P \times \frac{qA}{c} + \frac{qA}{c} \times \frac{qA}{c} \\ &= -\frac{q}{c} \left(A \times P + P \times A \right) \\ &= \frac{iq\hbar}{c} \left(A \times \nabla + \nabla \times A \right) \end{aligned}$$

The simplest way to manipulate operators is to act on a test function ψ

$$\begin{aligned} \frac{iq\hbar}{c} \left(A \times \nabla + \nabla \times A \right) \psi &= \frac{iq\hbar}{c} \left(A \times \nabla \psi + \nabla \times (\psi A) \right) \\ &= \frac{iq\hbar}{c} \left(A \times \nabla \psi + (\nabla \psi) A + (\nabla \times A) \psi \right) \\ &= \frac{iq\hbar}{c} \left(\nabla \times A \right) \psi \\ &= \frac{iq\hbar}{c} B \psi \end{aligned}$$

Therefore

$$\pi \times \pi = \frac{iq\hbar}{c} B$$

Shankar 20.1.1

Solve for the 4 spinors w that satisfy Shankar Eq. 20.3.3. You may assume that the 3- momentum $\vec{p}$ is along the z-axis. Normalize them to unity, and show that they are mutually orthogonal.

Equation (20.3.3) is

$$Ew = (\alpha \cdot p + \beta m)w$$

In terms of the relativistic and non-relativistic components,

$$\begin{pmatrix} E-m & -\sigma \cdot p \\ -\sigma \cdot p & E+m \end{pmatrix} \begin{pmatrix} \chi \\ \Phi \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The solutions are given in equations (20.3.7) and (20.3.8). Choosing the basis $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ for Φ and letting p be in the z direction,

$$w_{1,3} = \begin{pmatrix} p/(\pm E - m) \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad w_{2,4} = \begin{pmatrix} 0 \\ p/(\pm E - m) \\ \\ 0 \\ 1 \end{pmatrix}$$

Orthogonality of the spinors is easy to see using $E^2 = p^2 + m^2$.

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The five terms in 20.2.28 are

$$\begin{array}{rl} \dfrac{P^2}{2m} & \text{Hermitian} \\ V & \text{Hermitian} \\ -\dfrac{P^4}{8m^3c^2} & \text{Hermitian} \\ \dfrac{i\sigma \cdot P \times [P,V]}{4m^2c^2} & \text{Hermitian} \\ \dfrac{P \cdot [P,V]}{4m^2c^2} & \text{anti-Hermitian} \end{array}$$

Recall that $\hat{P}$ is a Hermitian operator, the potential V is assumed to be Hermitian (for conservation of probability) and the Pauli matrices σ are Hermitian. The commutator $[X,Y]$ of two Hermitian operators is anti-Hermitian since $[X,Y]^\dagger = (XY)^\dagger - (YX)^\dagger = Y^\dagger X^\dagger - X^\dagger Y^\dagger = YX - XY = -[X,Y]$. The cross product of two Hermitian (vector) operators is again Hermitian since $(X \times Y)^\dagger = X^\dagger \times Y^\dagger = X \times Y$.